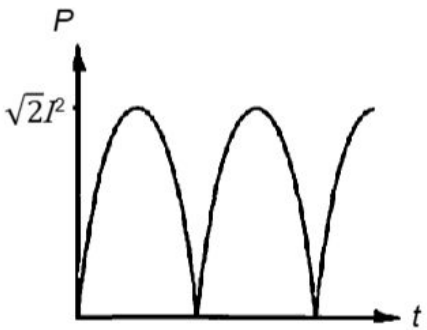
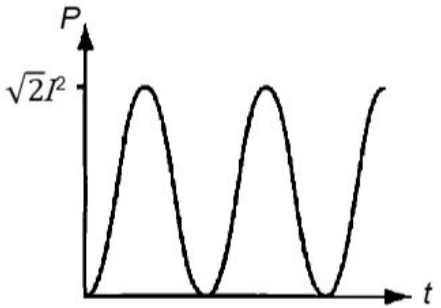
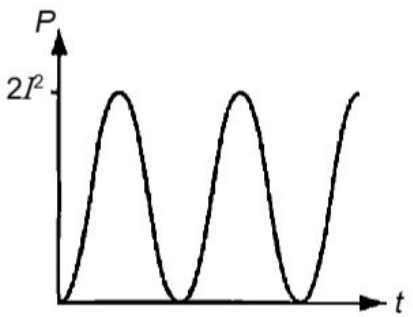
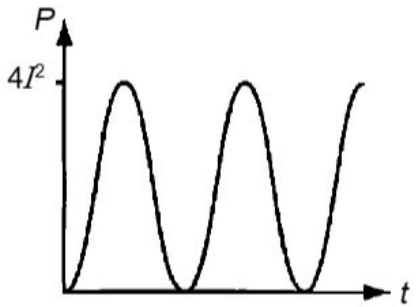
24	A sinusoidal alternating current of root-mean-square (r.m.s.) value I is passed through a resistor of resistance 2.0 Ω. Which graph shows how the power P dissipated in the resistor varies with time t?

[Turn over

	A
	
	B
	
	C
	
	D
	
L1	**Answer: D** Instantaneous power for sinusoidal a.c. $P = [I_o \sin(\omega t)]^2 \times R = I_o^2 R \sin^2(\omega t)$ → Answer is either B, C or D since this is a sine-squared graph. Mean power, $\langle P \rangle = I_{rms}^2 R$ Since this is a sinusoidal a.c., $P_o = 2\langle P \rangle = 2 I_{rms}^2 R = 2 I^2 (2) = 4 I^2$